

DSA (Abdul Bari) - Day 3

45-day plan | YouTube playlist course

Asymptotic Notations

Big-O, Omega and Theta - upper, lower and tight bounds.

Key concepts

- O = upper bound (worst), Ω = lower bound (best), Θ = tight bound.
- Analyse best/worst/average cases.
- Amortised analysis for occasional costly ops.

Key idea / example

Linear search: best $O(1)$, worst $O(n)$, average $O(n)$.

Today's tasks (1.5h)

- ☐ Define $O/\Omega/\Theta$
- ☐ Give best & worst of linear search
- ☐ Note amortised cost

Resources

Playlist: youtube.com/playlist?list=PLsr8vTgyLdy_YndxNcI4WkH5Vorj5qvrV

(Clickable playlist link is in the app: Goals > this day > Study block.)